Assessing Laboratory Testing Capacity in the BDBV outbreak, DRC

Report #2 – Published 4 June 2026 - Data as of 1 June 2026

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Executive summary

This report assesses laboratory testing capacity during the early phase of the Bundibugyo virus disease outbreak in the DRC, declared on 15 May 2026. Using data compiled from 18 INSP sitreps covering 14 May to 1 June, we examine what proportion of suspected cases were sampled, how rapidly samples moved through the laboratory process, and where delays persisted. Overall, the findings suggest that laboratory testing capacity improved over the past 2 weeks, but important bottlenecks remained, particularly in sample shipment.

- By 29 May, 82.7% of suspected cases had a sample received at the laboratory, and 71.6% had a sample tested, up from 5.3% for both indicators on 14 May.
- Positivity among tested samples fell from 61.5% to 32.0%, consistent with testing expanding beyond the highest-suspicion cases.
- The average time to diagnosis decreased from 10 days on 14 May to 7.5 days by 24 May, but shipment delays remained substantial at 4.5 days on 29 May.
- Within-laboratory delay averaged 1.8 days but fluctuated over time, suggesting limited reserve capacity and vulnerability to surges.
- Interpretation remains constrained by limitations in the situation reports, including incomplete laboratory detail, inconsistent reporting across sitreps, and uncertainty about how tested samples and results were classified.

Introduction

In the two weeks following the declaration of the BDBV outbreak in DRC on 15 May 2026, many suspected cases were reported, but only a small number were confirmed. This reflects major constraints in diagnostic capacity for Bundibugyo virus, which are slowing the response and limiting visibility on the true scale of the outbreak.

Over the weekend of 29 May, suspected cases were substantially reclassified, causing a sharp drop in their number and a rise in confirmed cases. This suggests that diagnostic capacity is beginning to catch up with suspected case reporting. However, laboratory data remain limited to daily INSP sitreps, which are sometimes difficult to interpret.

Here, we combine data from 18 sitreps to quantify several indicators of laboratory testing capacity between 14 May and 1 June. Although the sitreps usually combine data from the Bunia, Kinshasa, Beni, Goma, and Butembo laboratories, 2 sitreps (Nos. 13 and 15) included Bunia only. Our analysis pools all available laboratory data and does not assess individual laboratories separately.

Our analysis focuses on the journey of a sample; from the time it is taken from a suspected case until a positive result is obtained (figure 1).

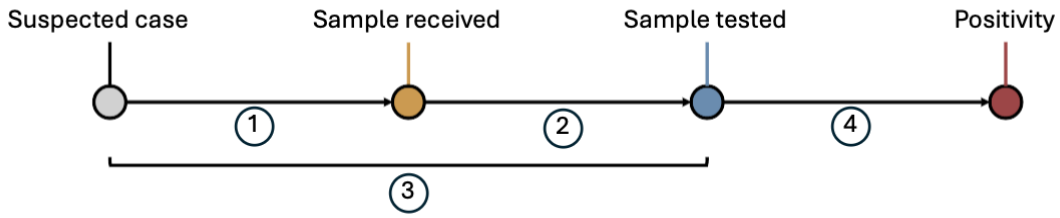


Figure 1. *Sample journey: a sample is collected from a suspected case and sent to the laboratory. Once received, it is stored and then tested by PCR, ultimately yielding either a positive or negative result. The numbers refer to the performance indicators defined below.*

Definitions

We define 7 indicators to assess laboratory performance.

Proportions indicators

1. **Proportion of suspected case with a sample received at the lab,** calculated**:**
   N samples received / N suspected cases
2. **Proportion of samples at the lab already tested,** calculated**:**
   N samples tested / N samples received
3. **Proportion of suspected cases with a sample already tested**, calculated**:**
   N samples tested / N suspected cases
4. **Proportion of positive samples among those tested (positivity)**. This indicator is only meaningful when considered alongside the proportion of suspected cases that were sampled, because high positivity may reflect either intense transmission or overly restrictive testing. It is calculated**:**
   N samples positives / N samples tested

Delay indicators

1. **Shipment delay**: time from notification of a new suspected case to receipt of the sample at the laboratory. As far as we understand, suspected cases are not always sampled immediately after notification, labs are sending sample collection teams to collect samples and transport back to the lab. Calculated from the cumulative number of new suspected cases and the cumulative number of samples received at the laboratory (see below).

2. **Within-laboratory delay**: time from laboratory receipt to sample testing. Calculated from the cumulative number of samples received at the laboratory and the cumulative number of samples tested (see below).

3. **Overall time-to-diagnosis delay**: time from sample collection in a suspected case to laboratory testing. Calculated from the cumulative number of new suspected cases and the cumulative number of samples tested (see below).

Methods

Data sources

Cumulative laboratory data were manually compiled and reconstructed from the daily INSP situation reports covering 14 May to 1 June.

Incidence data for suspected cases came from Epicentre's compiled dataset up to 25 May, and from the INSP GitHub repository thereafter (see our report #1 for more details). For indicators based on suspected cases, we report results from 14 to 29 May, the last date with reliable data on new suspected cases. After 29 May, only total suspected case counts were reported, making it impossible to separate newly suspected cases from cases removed after receiving a laboratory result.

Cumulative distribution and delays

Cumulative delays are estimated by comparing the cumulative curves for successive steps in the sample journey. For a given cumulative count on the upstream curve, we identify the date when the downstream curve reaches the same count; the difference between those dates is the delay in days. For example, if the cumulative number of new suspected cases reaches 100 on day D1 and the cumulative number of samples received reaches 100 on day D2, then the shipment delay at D1 is D2 − D1. In practice, we used interpolation to estimate the corresponding dates on the downstream curve.

Results

- The proportion of suspected cases with a sample received at the laboratory increased from 5.3% on 14 May to 82.7% by 29 May (Figure 2).
- A similar increase was observed in the proportion of suspected cases with a sample tested, rising from 5.3% on 14 May to 71.6% by 29 May (Figure 2).

➔ ***Together, these findings suggest that laboratories are catching up, with a growing share of suspected cases being sampled and tested.***

- The proportion of positive samples among those tested (positivity) decreased from 61.5% on 14 May to 32.0% by 29 May.

➔ ***This is consistent with testing expanding beyond the highest-suspicion cases, but it should be interpreted in light of the sampling strategy, which is unknown (e.g. prioritisation of high-risk contacts, etc).***

- The time-to-diagnosis delay decreased from 10 days on 14 May to 7.5 days by 24 May (Figure 3).
- This delay was driven mainly by shipment time, which remained 4.5 days on 29 May despite falling from 7 days on 14 May (Figure 3).

➔ ***This suggests that diagnostic delays remain substantial and are driven mainly by slow sample shipment. Several situation reports note samples awaiting transfer to laboratory facilities.***

- The proportion of samples received at the laboratory that were tested increased after 14 May, but remained variable, with some days showing only 50% of received samples tested (Figure 2).
- The average within-laboratory delay was 1.8 days, but fluctuated over the course of the outbreak. It reached 3 days on 21 May, fell to 1 day on 25 May, and began increasing again by 29 May (Figure 3; Appendix Figure 3).

➔ ***This suggests that laboratory capacity may be operating near its limit, leaving the workflow fragile and vulnerable to surges.***

Table 1. Summary of data available as of 29th May 2026

	Value as of 29 May 2026	**Source**
Data		
A: N suspected cases	1152	EpiDS Report #1
B: N samples received at the lab	953	Sitrep 15
C: N samples tested	825	Sitrep 15
D: N samples positive	264	Sitrep 15
Proportion indicators		
1: Proportion of suspected case with a sample received at the lab	82.7%	=B/A
2: Proportion of samples at the lab already tested	86.6%	=C/B
3: Proportion of suspected cases with a sample already tested	71.6%	=C/A
4: Proportion of positive samples among those tested (positivity)	32.0%	=D/C
Delay indicators		
1: Average shipment delay (period: 14 to 27 May)	5.9 days	Calculated from cumulative curves of A and B
2: Average within-laboratory delay (period 14 to 29 May)	1.8 days	Calculated from cumulative curves of B and C
3: Average time-to-diagnosis delay (period 14 to 24 May)	8.6 days	Calculated from cumulative curves of A and C

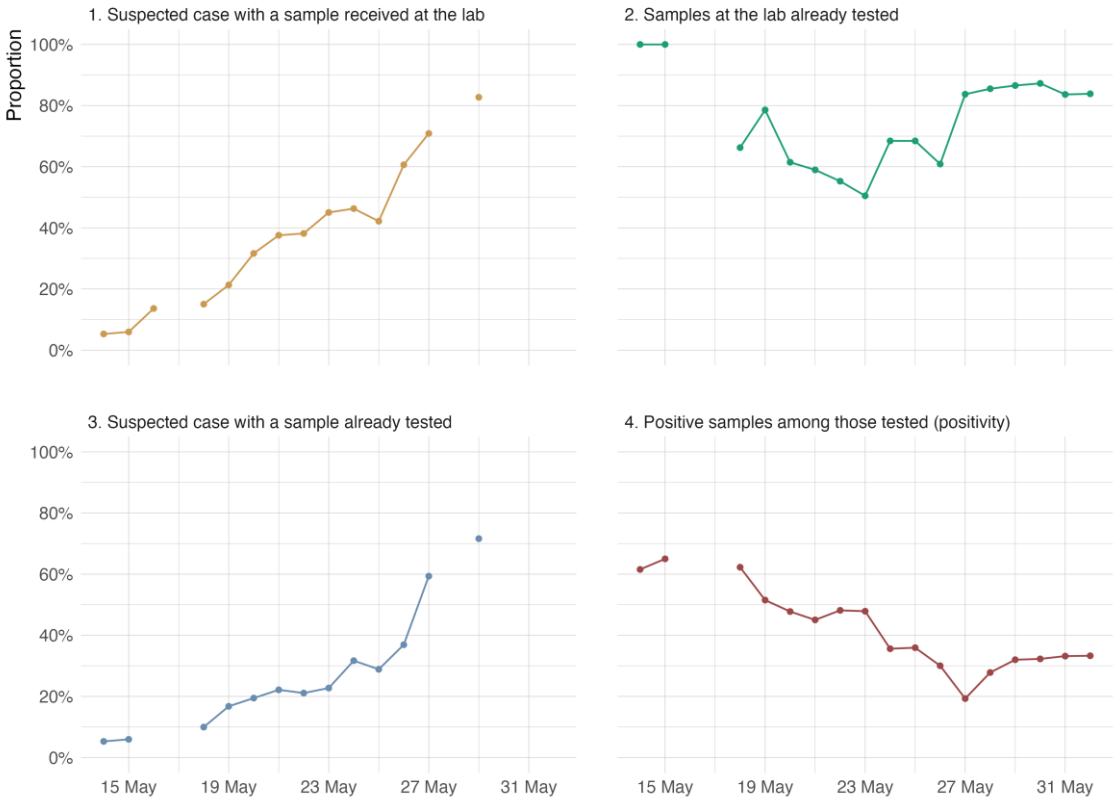


Figure 2. *Proportion indicators over time. See the definitions above for more details.*

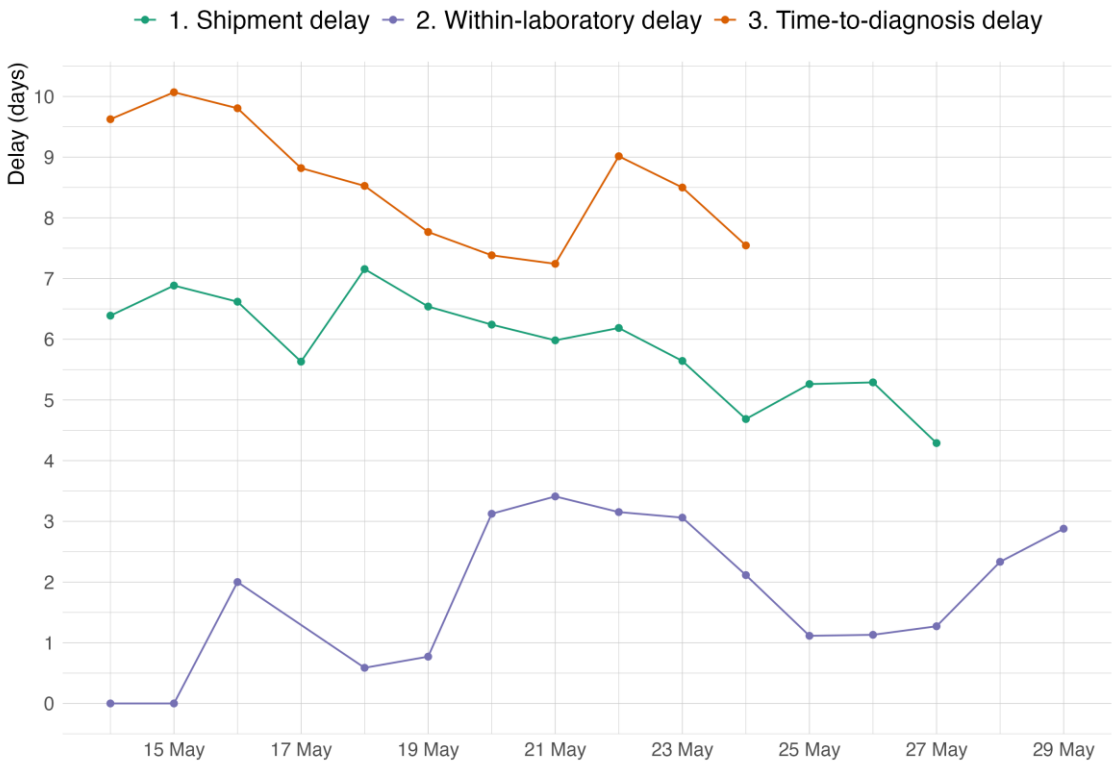


Figure 3. *Delay indicators over time. See the definitions above for more details.*

Limitations

There are several limitations that must be considered when interpreting the data

- We assumed that all tested samples have a valid result (positive or negative, not inconclusive), but this is unclear from the sitreps. If this assumption is incorrect, the denominator used to calculate positivity is inflated (e.g. we are underestimating positivity).

- Data by health zone are unreliable up to sitrep 10. From that point onward, only cumulative case totals are reported, so we do not present analyses stratified by health zone.

- Sitreps 13 and 15 provide laboratory data for Bunia only.

- Report 3 (16 May) does not include the number of tested samples, and report 11 (25 May) appears unreliable because its figures are identical to those of the previous sitrep.

Remaining questions

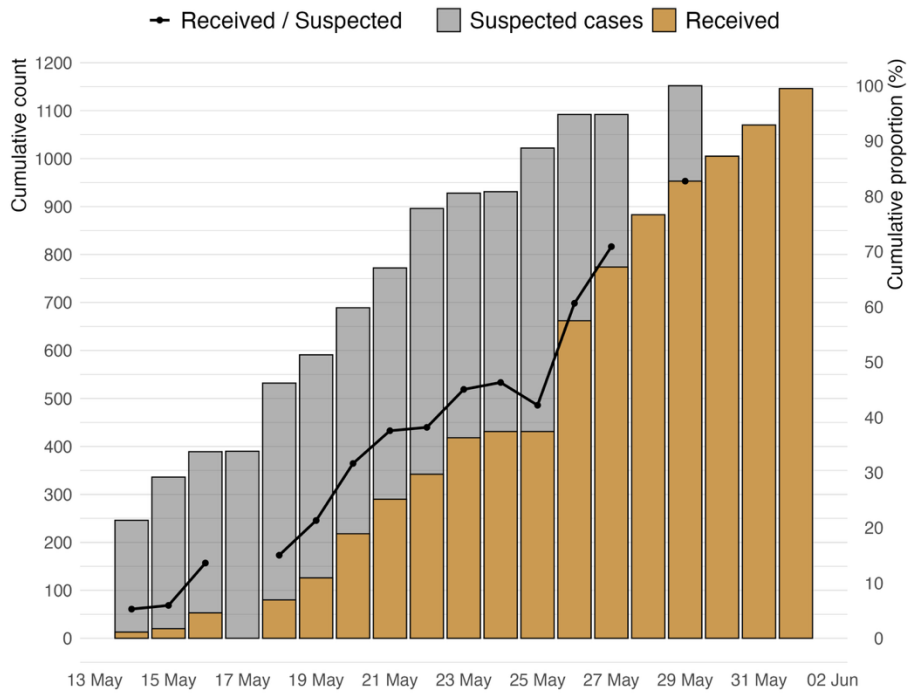
Several important questions about the laboratory diagnostic remain and will require more detailed data:

1. What proportion of suspected cases are sampled each day?
2. What criteria guide the sampling strategy—that is, who gets tested?
3. Sitreps describe samples as "tested," but it is unclear what this includes. How are negative, positive, inconclusive, and pending results recorded?

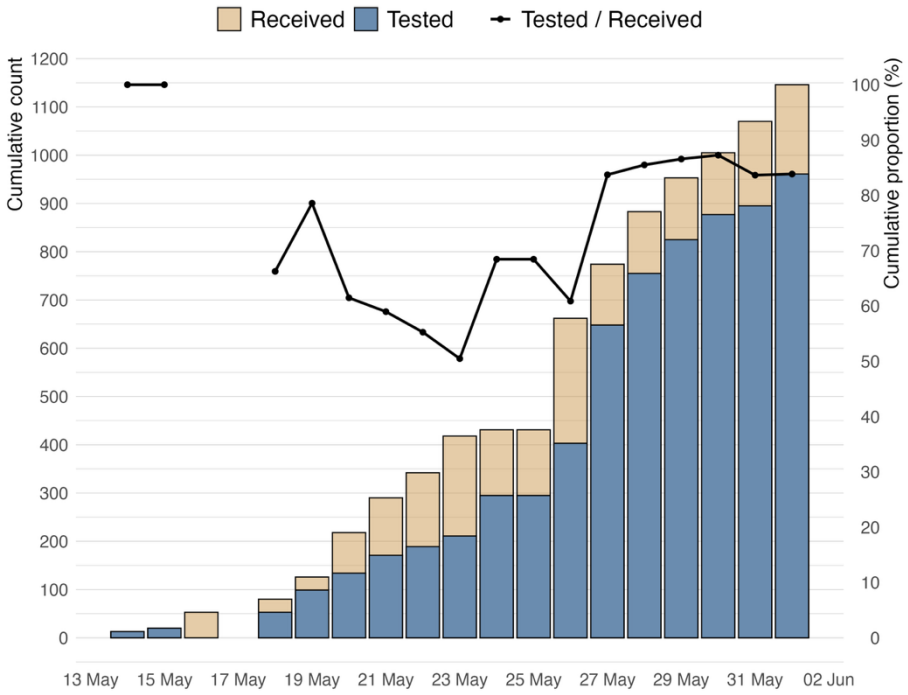
Acknowledgments

Patrick Barks, Clara Brigitta, Paul Campbell, Maelenn Gouillou, Henry Laurenson-Schafer, Mathilde Mousset, Erica Simons

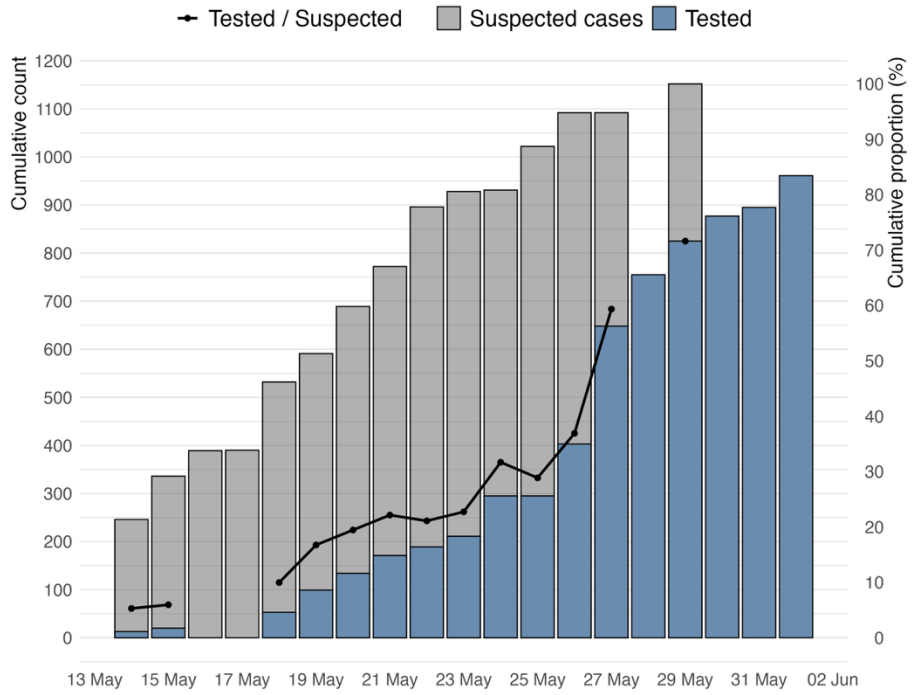
Appendix



Appendix Figure 1. *Cumulative numbers of suspected cases (grey bars), samples received (orange bars), and proportion of suspected case with a sample received at the lab (black line).*

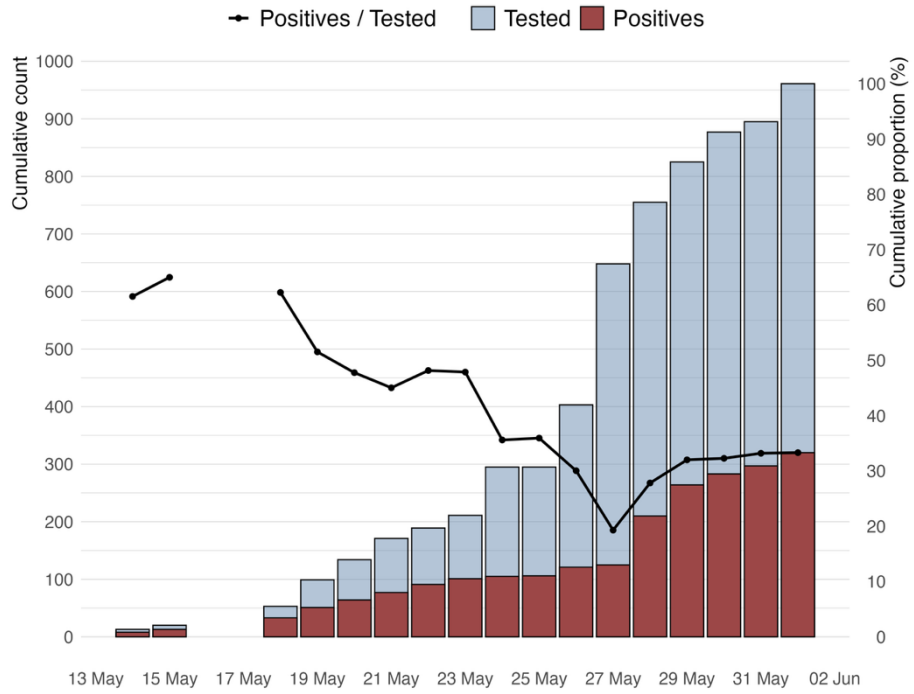


Appendix Figure 2. *Cumulative numbers of samples received (yellow bars), samples tested (blue bars), and proportion of samples at the lab already tested (black line).*



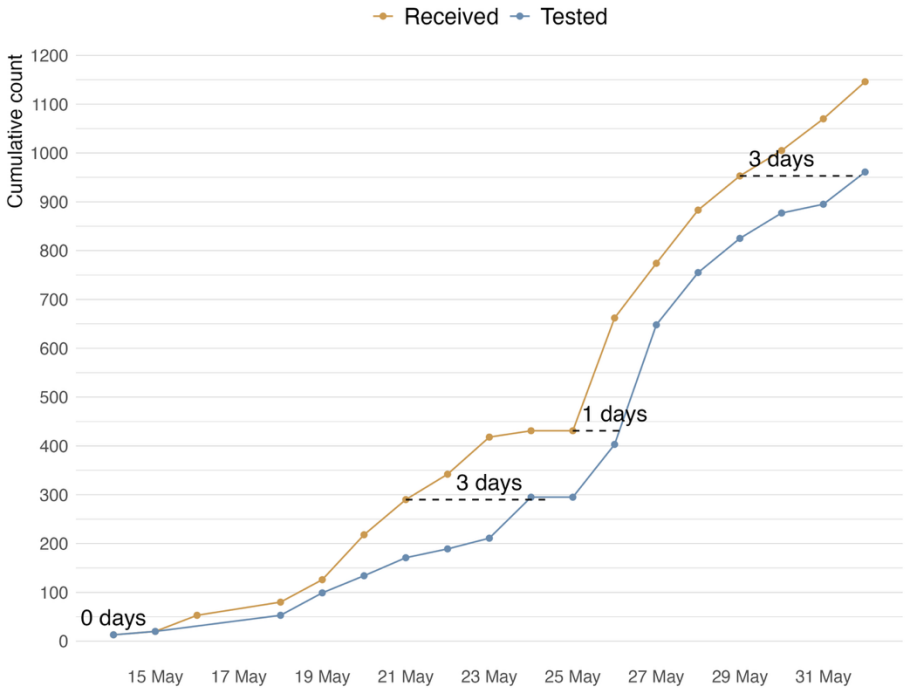
***Appendix Figure 3**. Cumulative numbers of suspected cases (grey bars), samples tested (blue bars), and proportion of suspected case with a sample already tested (black line).*

***Appendix Figure 4.** Cumulative numbers of samples tested (blue bars), samples tested positives (red bars), and proportion of samples tested that are positives (black line).*

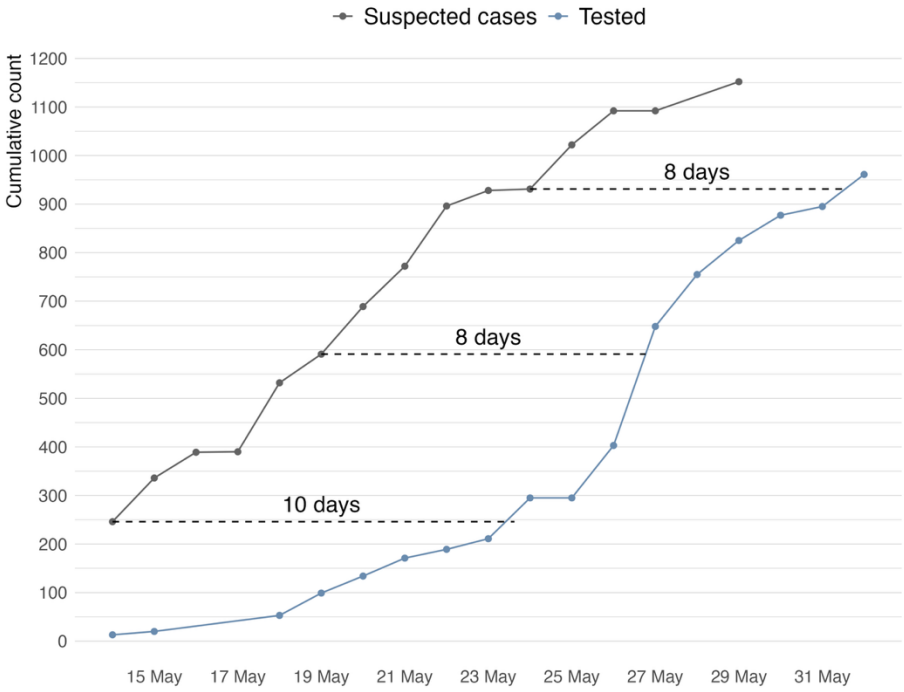




Appendix Figure 5. *Cumulative counts of suspected cases and received samples. The delay between curves is the time taken for received samples to reach a given level of suspected cases,* ***on average 5.9 days****.*



Appendix Figure 6. *Cumulative counts of samples received and samples tested. The delay between curves is the time taken for received samples to be tested* ***on average 1.8 days****.*



Appendix Figure 7. *Cumulative counts of suspected cases and samples tested. The delay between curves is the time taken for suspected cases to be tested,* ***on average 8.6 days****.*